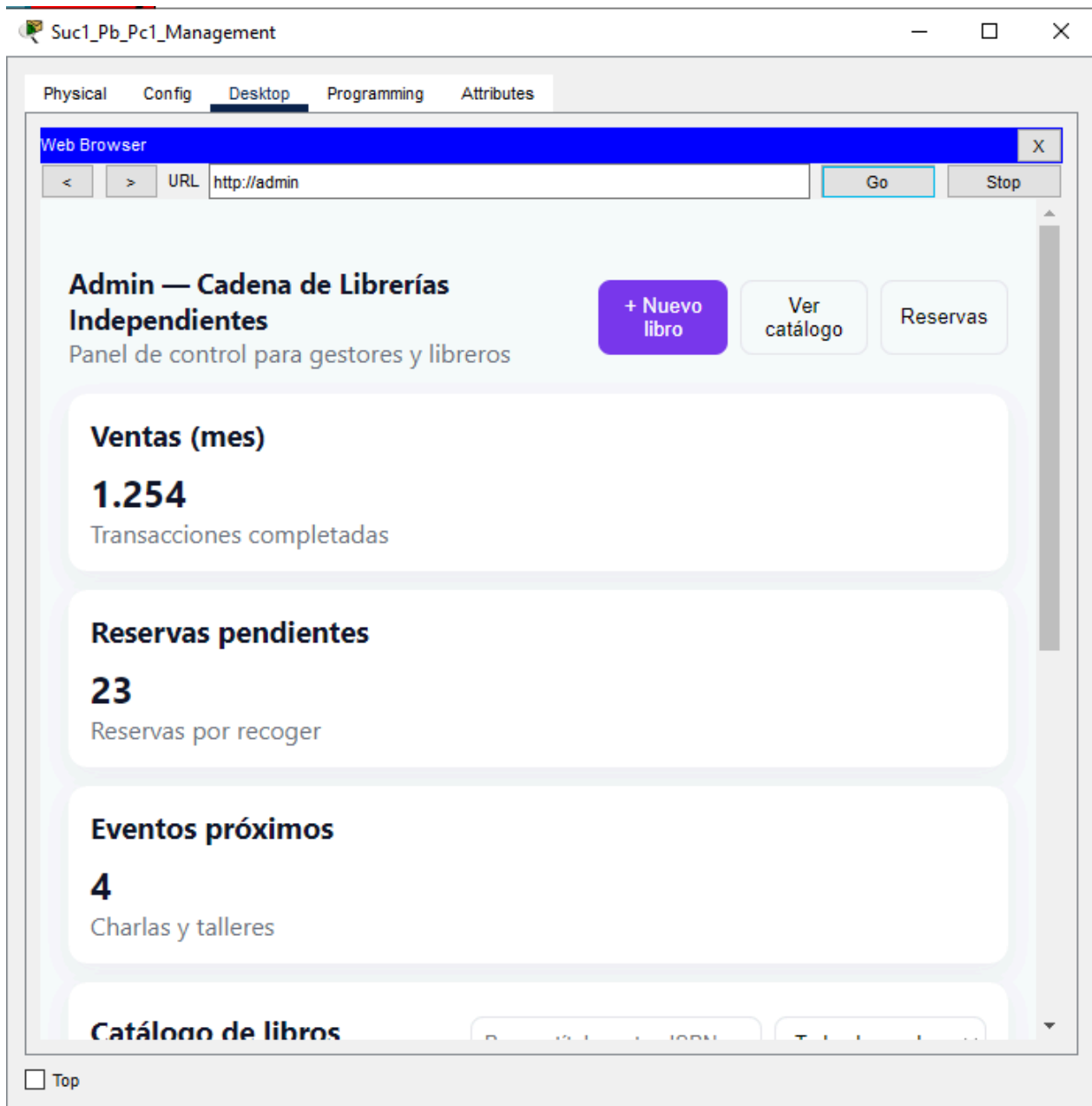
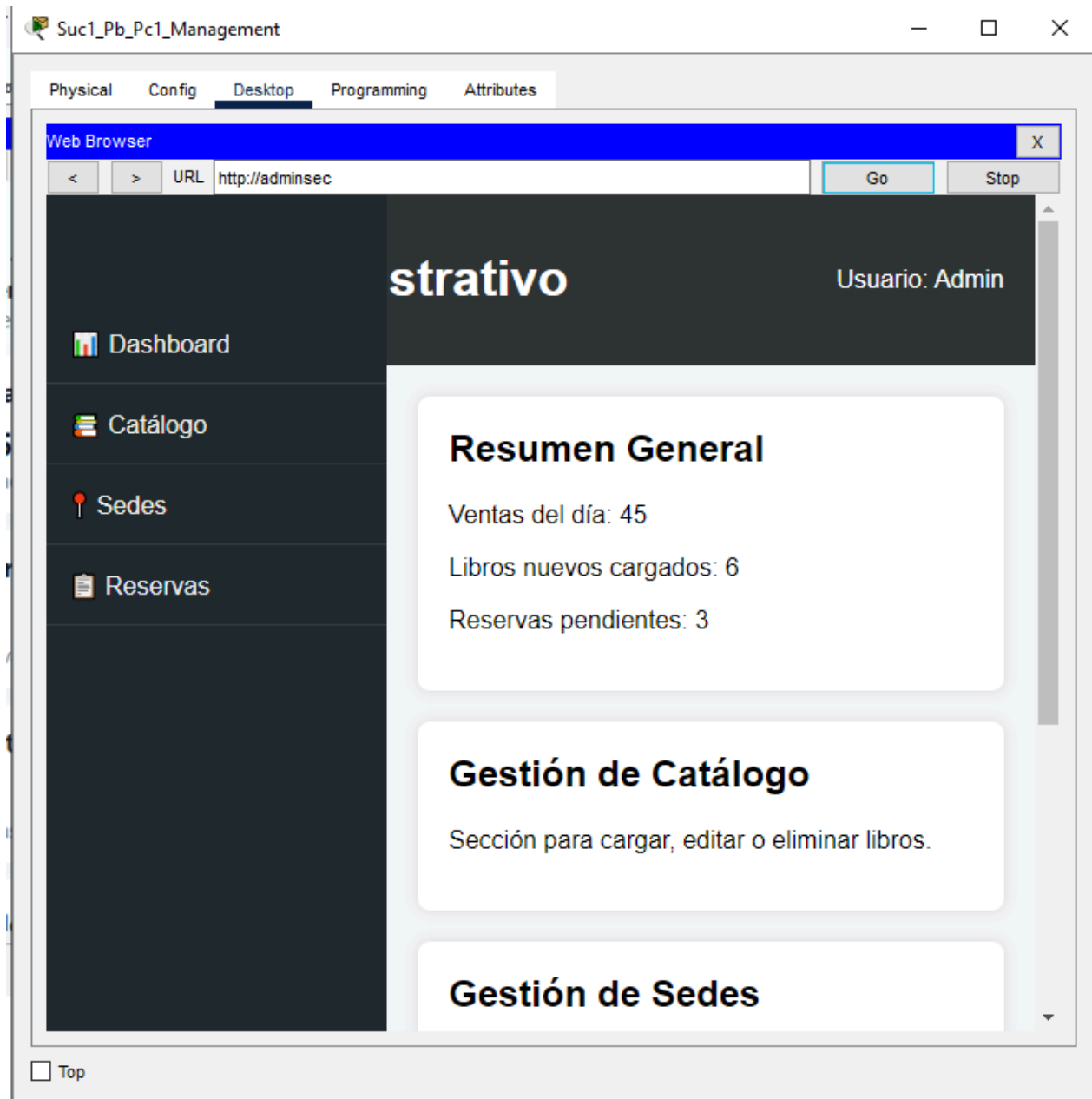


Evidencias

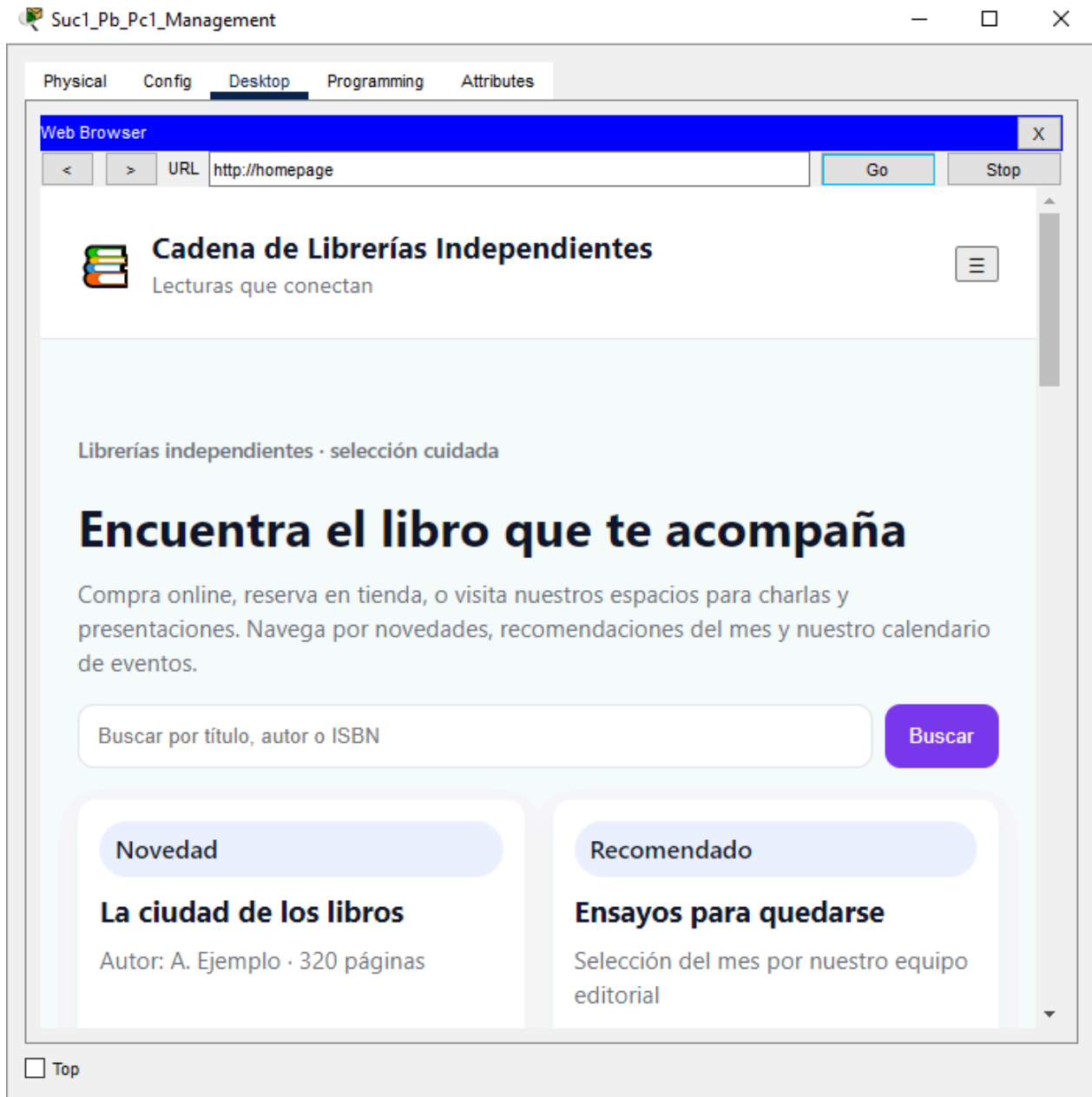
Servicio Web Administrativa S1 Funcional



Servicio Web Administrativa Secundario S1 Funcional



Página Principal S1 **Funcional**



FLSM Sucursal 1

```

Address:      172.20.10.0      10101100.00010100.00001010. 00000000
Netmask:      255.255.255.0 = 24 11111111.11111111.11111111. 00000000
Wildcard:     0.0.0.255      00000000.00000000.00000000. 11111111
=>
Network:      172.20.10.0/24   10101100.00010100.00001010. 00000000
HostMin:      172.20.10.1      10101100.00010100.00001010. 00000001
HostMax:      172.20.10.254    10101100.00010100.00001010. 11111110
Broadcast:    172.20.10.255    10101100.00010100.00001010. 11111111
Hosts/Net:    254              Class B, Private Internet

Subnets after transition from /24 to /27
Netmask:      255.255.255.224 = 27 11111111.11111111.11111111.111 000000
Wildcard:     0.0.0.31        00000000.00000000.00000000.000 11111
1.
Network:      172.20.10.0/27   10101100.00010100.00001010.000 000000
HostMin:      172.20.10.1      10101100.00010100.00001010.000 000001
HostMax:      172.20.10.30     10101100.00010100.00001010.000 111110
Broadcast:    172.20.10.31     10101100.00010100.00001010.000 111111
Hosts/Net:    30              Class B, Private Internet
2.
Network:      172.20.10.32/27  10101100.00010100.00001010.001 000000
HostMin:      172.20.10.33     10101100.00010100.00001010.001 000001
HostMax:      172.20.10.62     10101100.00010100.00001010.001 111110
Broadcast:    172.20.10.63     10101100.00010100.00001010.001 111111
Hosts/Net:    30              Class B, Private Internet
3.
Network:      172.20.10.64/27  10101100.00010100.00001010.010 000000
HostMin:      172.20.10.65     10101100.00010100.00001010.010 000001
HostMax:      172.20.10.94     10101100.00010100.00001010.010 111110
Broadcast:    172.20.10.95     10101100.00010100.00001010.010 111111
Hosts/Net:    30              Class B, Private Internet
4.
Network:      172.20.10.96/27  10101100.00010100.00001010.011 000000
HostMin:      172.20.10.97     10101100.00010100.00001010.011 000001
HostMax:      172.20.10.126    10101100.00010100.00001010.011 111110
Broadcast:    172.20.10.127    10101100.00010100.00001010.011 111111
Hosts/Net:    30              Class B, Private Internet
5.
Network:      172.20.10.128/27 10101100.00010100.00001010.100 000000
HostMin:      172.20.10.129    10101100.00010100.00001010.100 000001
HostMax:      172.20.10.158    10101100.00010100.00001010.100 111110
Broadcast:    172.20.10.159    10101100.00010100.00001010.100 111111
Hosts/Net:    30              Class B, Private Internet

```

VLSM Sucursal 2

The network 192.168.50.0/24 has 254 hosts. Your subnets need 97 hosts.								
NAME	HOSTS NEEDED	HOSTS AVAILABLE	UNUSED HOSTS	NETWORK ADDRESS	SLASH	MASK	USABLE RANGE	BROADCAST
2	40	62	22	192.168.50.0	/26	255.255.255.192	192.168.50.1 - 192.168.50.62	192.168.50.63
3	25	30	5	192.168.50.64	/27	255.255.255.224	192.168.50.65 - 192.168.50.94	192.168.50.95
5	20	30	10	192.168.50.96	/27	255.255.255.224	192.168.50.97 - 192.168.50.126	192.168.50.127
1	8	14	6	192.168.50.128	/28	255.255.255.240	192.168.50.129 - 192.168.50.142	192.168.50.143
4	4	6	2	192.168.50.144	/29	255.255.255.248	192.168.50.145 - 192.168.50.150	192.168.50.151

FLSM Sucursal 3

```

Address:      172.68.0.0          10101100.01000100.00000000. 00000000
Netmask:      255.255.255.0 = 24  11111111.11111111.11111111. 00000000
Wildcard:     0.0.0.255          00000000.00000000.00000000. 11111111
=>
Network:      172.68.0.0/24      10101100.01000100.00000000. 00000000
HostMin:      172.68.0.1         10101100.01000100.00000000. 00000001
HostMax:      172.68.0.254       10101100.01000100.00000000. 11111110
Broadcast:    172.68.0.255       10101100.01000100.00000000. 11111111
Hosts/Net:    254                Class B
Subnets after transition from /24 to /27
Netmask:      255.255.255.224 = 27 11111111.11111111.11111111.111 000000
Wildcard:     0.0.0.31          00000000.00000000.00000000.000 111111
1.
Network:      172.68.0.0/27      10101100.01000100.00000000.000 000000
HostMin:      172.68.0.1         10101100.01000100.00000000.000 000001
HostMax:      172.68.0.30       10101100.01000100.00000000.000 111110
Broadcast:    172.68.0.31       10101100.01000100.00000000.000 111111
Hosts/Net:    30                Class B
2.
Network:      172.68.0.32/27     10101100.01000100.00000000.001 000000
HostMin:      172.68.0.33       10101100.01000100.00000000.001 000001
HostMax:      172.68.0.62       10101100.01000100.00000000.001 111110
Broadcast:    172.68.0.63       10101100.01000100.00000000.001 111111
Hosts/Net:    30                Class B
3.
Network:      172.68.0.64/27     10101100.01000100.00000000.010 000000
HostMin:      172.68.0.65       10101100.01000100.00000000.010 000001
HostMax:      172.68.0.94       10101100.01000100.00000000.010 111110
Broadcast:    172.68.0.95       10101100.01000100.00000000.010 111111
Hosts/Net:    30                Class B
4.
Network:      172.68.0.96/27     10101100.01000100.00000000.011 000000
HostMin:      172.68.0.97       10101100.01000100.00000000.011 000001
HostMax:      172.68.0.126      10101100.01000100.00000000.011 111110
Broadcast:    172.68.0.127      10101100.01000100.00000000.011 111111
Hosts/Net:    30                Class B
5.
Network:      172.68.0.128/27    10101100.01000100.00000000.100 000000
HostMin:      172.68.0.129      10101100.01000100.00000000.100 000001
HostMax:      172.68.0.158      10101100.01000100.00000000.100 111110
Broadcast:    172.68.0.159      10101100.01000100.00000000.100 111111
Hosts/Net:    30                Class B

```

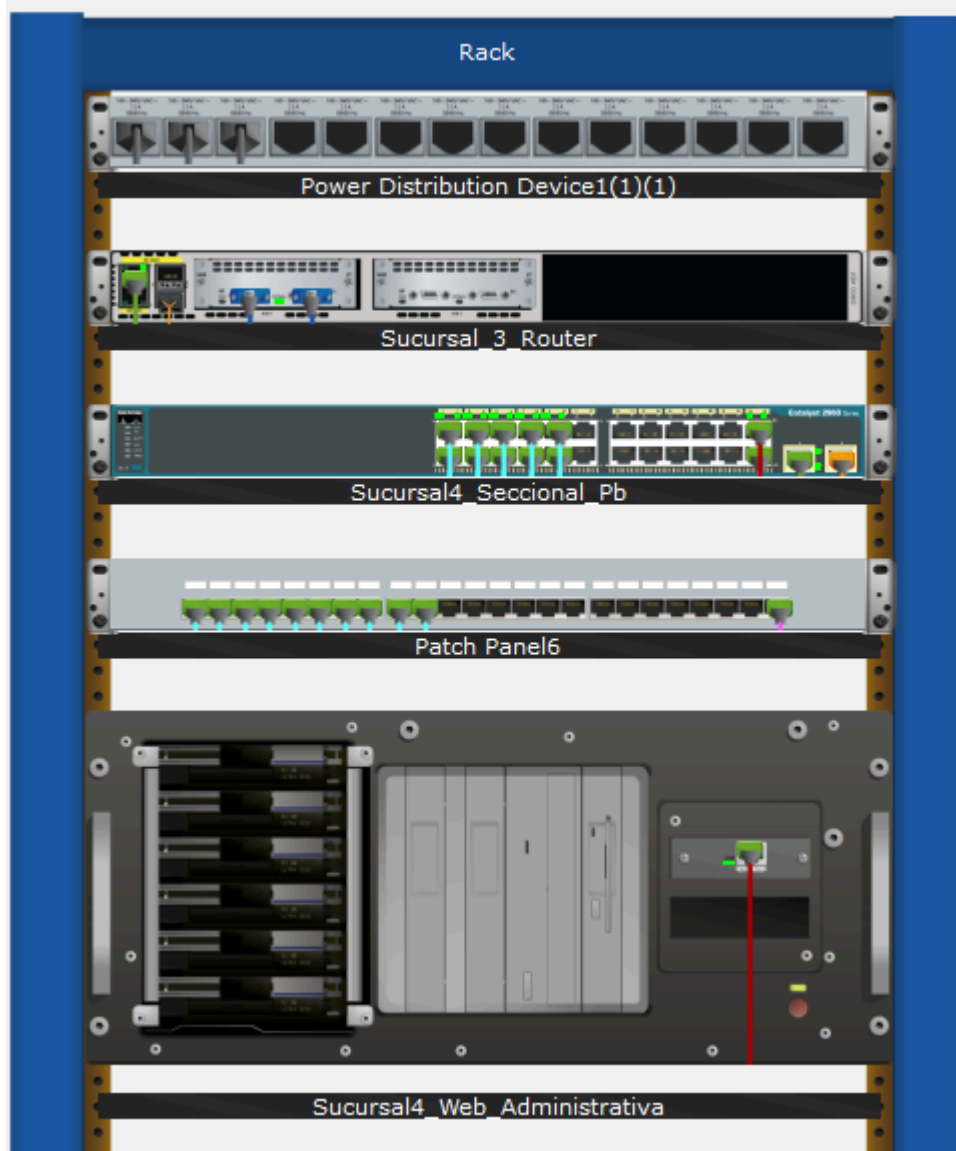
FLSM Sucursal 4

```

Address:      198.51.100.0      11000110.00110011.01100100. 00000000
Netmask:      255.255.255.0 = 24 11111111.11111111.11111111. 00000000
Wildcard:     0.0.0.255        00000000.00000000.00000000. 11111111
=>
Network:      198.51.100.0/24   11000110.00110011.01100100. 00000000
HostMin:      198.51.100.1      11000110.00110011.01100100. 00000001
HostMax:      198.51.100.254    11000110.00110011.01100100. 11111110
Broadcast:    198.51.100.255    11000110.00110011.01100100. 11111111
Hosts/Net:    254                Class C
Subnets after transition from /24 to /27
Netmask:      255.255.255.224 = 27 11111111.11111111.11111111.111 000000
Wildcard:     0.0.0.31          00000000.00000000.00000000.000 11111
1.
Network:      198.51.100.0/27   11000110.00110011.01100100.000 000000
HostMin:      198.51.100.1      11000110.00110011.01100100.000 000001
HostMax:      198.51.100.30     11000110.00110011.01100100.000 111110
Broadcast:    198.51.100.31     11000110.00110011.01100100.000 111111
Hosts/Net:    30                Class C
2.
Network:      198.51.100.32/27  11000110.00110011.01100100.001 000000
HostMin:      198.51.100.33     11000110.00110011.01100100.001 000001
HostMax:      198.51.100.62     11000110.00110011.01100100.001 111110
Broadcast:    198.51.100.63     11000110.00110011.01100100.001 111111
Hosts/Net:    30                Class C
3.
Network:      198.51.100.64/27  11000110.00110011.01100100.010 000000
HostMin:      198.51.100.65     11000110.00110011.01100100.010 000001
HostMax:      198.51.100.94     11000110.00110011.01100100.010 111110
Broadcast:    198.51.100.95     11000110.00110011.01100100.010 111111
Hosts/Net:    30                Class C
4.
Network:      198.51.100.96/27  11000110.00110011.01100100.011 000000
HostMin:      198.51.100.97     11000110.00110011.01100100.011 000001
HostMax:      198.51.100.126    11000110.00110011.01100100.011 111110
Broadcast:    198.51.100.127    11000110.00110011.01100100.011 111111
Hosts/Net:    30                Class C
5.
Network:      198.51.100.128/27 11000110.00110011.01100100.100 000000
HostMin:      198.51.100.129    11000110.00110011.01100100.100 000001
HostMax:      198.51.100.158    11000110.00110011.01100100.100 111110
Broadcast:    198.51.100.159    11000110.00110011.01100100.100 111111
Hosts/Net:    30                Class C

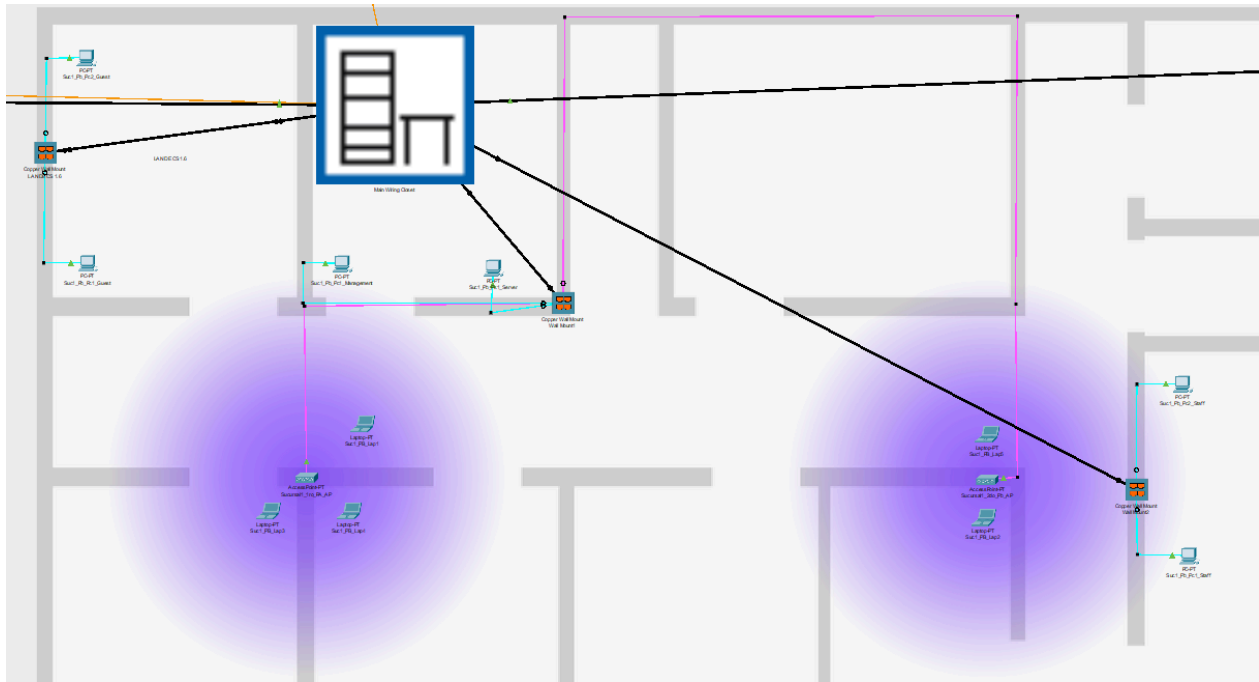
```

Rack de Ejemplo (Sucursal 4):



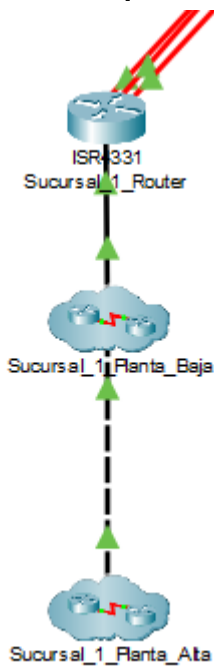
Cables separados por Servidor/PC/Access Point u otros.

Topología utilizada (Ejemplo Sucursal 1):

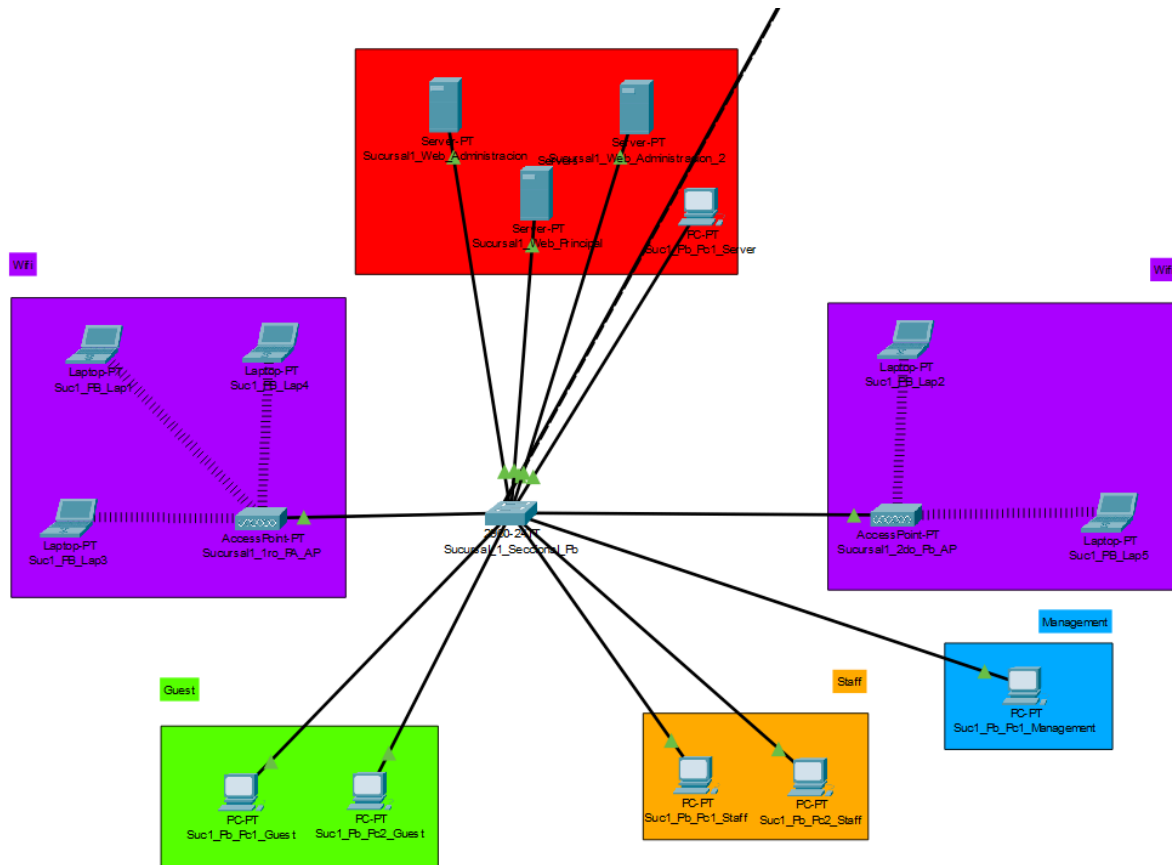


Algunas sucursales tienen los Access Point en un rack apartado del resto de dispositivos del rack del piso.

Cada sucursal se encuentra encapsulada en un Clúster que a su vez se divide en dos Clúster más que vienen a representar primer y segundo piso.



Como se ve cada piso:



VPN S1-S2:

```
Router#show crypto ipsec sa
```

```
interface: Serial0/1/1
  Crypto map tag: S1-S2, local addr 5.0.0.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (172.20.10.0/255.255.255.0/0/0)
remote  ident (addr/mask/prot/port): (192.168.50.0/255.255.255.0/0/0)
current_peer 5.0.0.2 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 1, #pkts encrypt: 1, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 5.0.0.1, remote crypto endpt.:5.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/1
current outbound spi: 0x01F418CE(32774350)

inbound esp sas:
  spi: 0x89D1F40F(2312238095)
--More--
```

VPN S2-S1:

```
Router#show crypto ipsec sa

interface: Serial0/1/0
  Crypto map tag: S2-S1, local addr 5.0.0.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.50.0/255.255.255.31/0/0)
  remote ident (addr/mask/prot/port): (172.20.10.0/255.255.255.31/0/0)
  current_peer 5.0.0.1 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 2, #pkts decrypt: 2, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

    local crypto endpt.: 5.0.0.2, remote crypto endpt.: 5.0.0.1
    path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
    current outbound spi: 0x89D1F40F(2312238095)

  inbound esp sas:
    spi: 0x01F418CE(32774350)
--More-- |
```

VPN S2-S3:

```

interface: Serial0/1/1
  Crypto map tag: S2-S3, local addr 5.0.0.5

  protected vrf: (none)
  local ident (addr/mask/prot/port): (192.168.50.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (172.68.0.0/255.255.255.0/0/0)
  current_peer 5.0.0.6 port 500
    PERMIT, flags={origin_is_acl,}
  #pkts encaps: 2, #pkts encrypt: 2, #pkts digest: 0
  #pkts decaps: 1, #pkts decrypt: 1, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 1, #recv errors 0

  local crypto endpt.: 5.0.0.5, remote crypto endpt.:5.0.0.6
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/1
  current outbound spi: 0xDD2DBE81(3710762625)

  inbound esp sas:
    spi: 0x515971BE(1364816318)
      transform: esp-aes esp-sha-hmac ,
      in use settings = {Tunnel, }
      conn id: 2005, flow_id: FPGA:1, crypto map: S2-S3
      sa timing: remaining key lifetime (k/sec): (4525504/3590)
      IV size: 16 bytes
      replay detection support: N
      Status: ACTIVE

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

```

VPN S3-S2:

Router#show crypto ipsec sa

```

interface: Serial0/1/1
  Crypto map tag: S3-S2, local addr 5.0.0.6

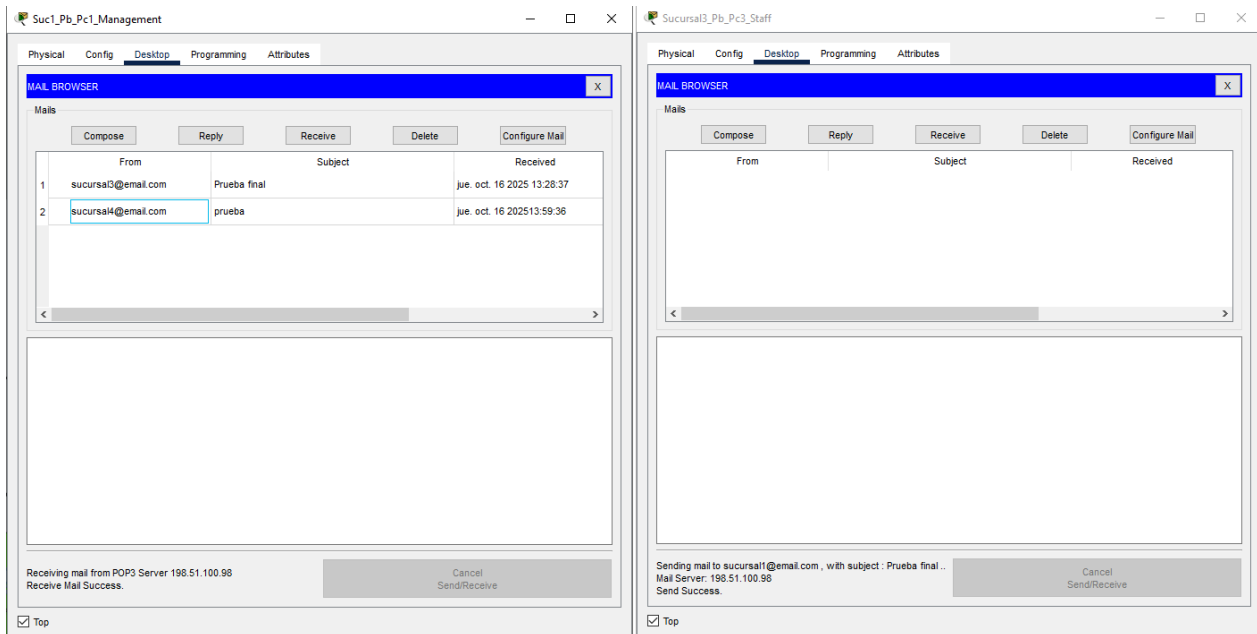
  protected vrf: (none)
  local ident (addr/mask/prot/port): (172.68.0.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.50.0/255.255.255.0/0/0)
  current_peer 5.0.0.5 port 500
    PERMIT, flags={origin_is_acl,}
  #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
  #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 5.0.0.6, remote crypto endpt.:5.0.0.5
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/1
  current outbound spi: 0x515971BE(1364816318)

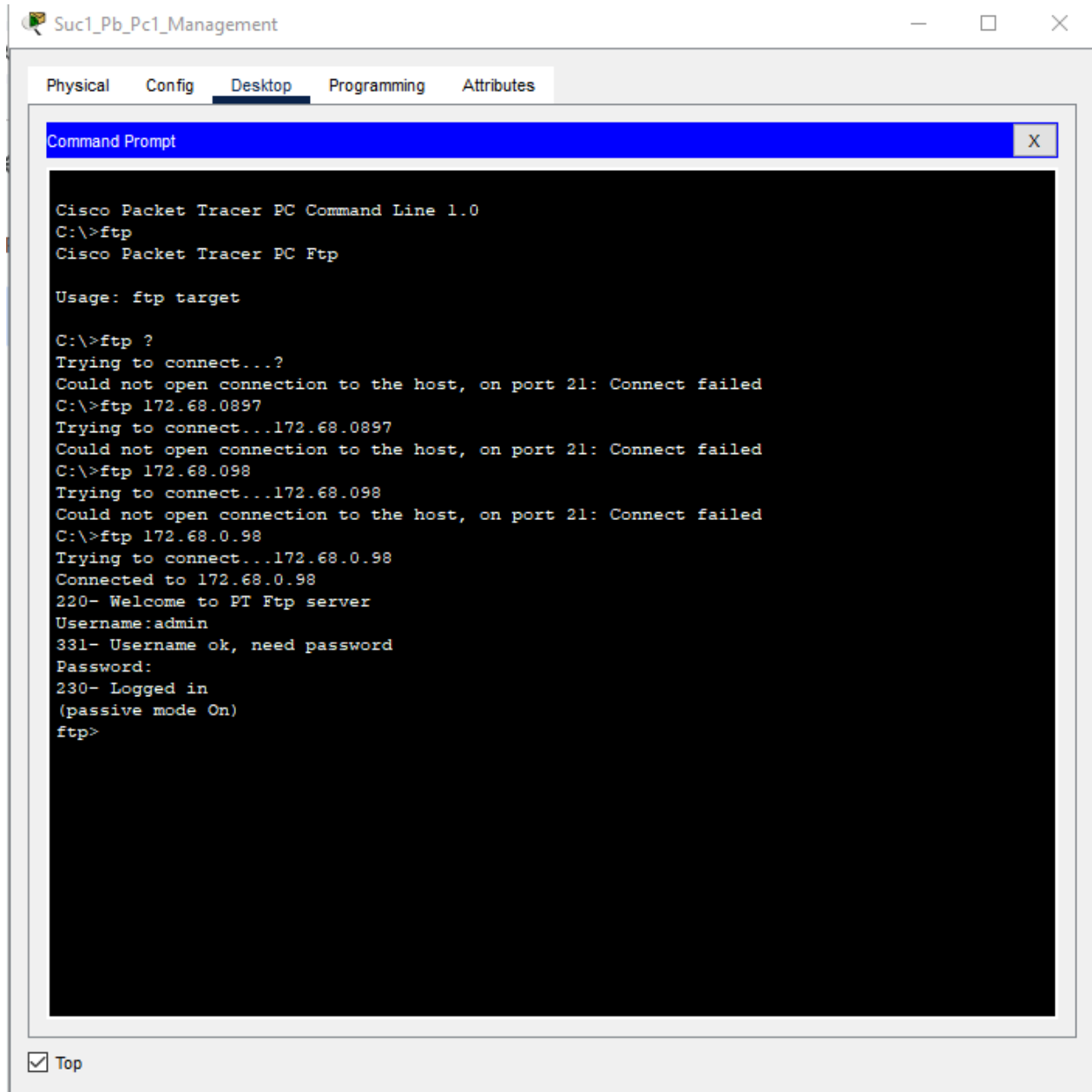
  inbound esp sas:
    spi: 0xDD2DBE81(3710762625)
--More-- |

```

SMTP Funcional:



FTP Funcional:



Es funcional; que yo tenga problemas al escribir una dirección IP es otra cosa 😂.

Comandos

Ejemplo de configuración del Router 1:

```
ip dhcp excluded-address 172.20.10.1
ip dhcp excluded-address 172.20.10.33
ip dhcp excluded-address 172.20.10.65
ip dhcp excluded-address 172.20.10.97
ip dhcp excluded-address 172.20.10.129
!
ip dhcp pool management_net
network 172.20.10.0 255.255.255.224
default-router 172.20.10.1
dns-server 8.8.8.2
ip dhcp pool staff_net
network 172.20.10.32 255.255.255.224
default-router 172.20.10.33
dns-server 8.8.8.2
ip dhcp pool wifi_net
network 172.20.10.64 255.255.255.224
default-router 172.20.10.65
dns-server 8.8.8.2
ip dhcp pool servers_net
network 172.20.10.96 255.255.255.224
default-router 172.20.10.97
dns-server 8.8.8.2
ip dhcp pool guest_net
network 172.20.10.128 255.255.255.224
default-router 172.20.10.129
dns-server 8.8.8.2
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
crypto isakmp policy 10
encr aes
authentication pre-share
group 2
!
crypto isakmp key admin address 5.0.0.2
!
crypto ipsec transform-set TSET esp-aes esp-sha-hmac
```

```
!  
crypto map S1-S2 10 ipsec-isakmp  
set peer 5.0.0.2  
set transform-set TSET  
match address 101  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface Tunnel1  
ip address 10.0.0.1 255.255.255.252  
mtu 1476  
tunnel source Serial0/1/1  
tunnel destination 5.0.0.2  
!  
!  
interface Tunnel3  
no ip address  
mtu 1476  
tunnel source Serial0/1/0  
tunnel destination 5.0.0.13  
!  
!  
interface Tunnel2  
ip address 10.0.0.5 255.255.255.252  
mtu 1476  
tunnel source GigabitEthernet0/0/2  
tunnel destination 6.0.0.9  
!  
!  
interface GigabitEthernet0/0/0  
no ip address  
duplex auto  
speed auto  
!
```

```
interface GigabitEthernet0/0/0.10
description management_net
encapsulation dot1Q 10
ip address 172.20.10.1 255.255.255.224
ip access-group ACL-S1-IN in
!
interface GigabitEthernet0/0/0.20
description staff_net
encapsulation dot1Q 20
ip address 172.20.10.33 255.255.255.224
ip access-group ACL-S1-IN in
!
interface GigabitEthernet0/0/0.30
description wifi_net
encapsulation dot1Q 30
ip address 172.20.10.65 255.255.255.224
ip access-group ACL-S1-IN in
!
interface GigabitEthernet0/0/0.40
description servers_net
encapsulation dot1Q 40
ip address 172.20.10.97 255.255.255.224
ip access-group ACL-S1-IN in
!
interface GigabitEthernet0/0/0.50
description guest_net
encapsulation dot1Q 50
ip address 172.20.10.129 255.255.255.224
ip access-group ACL-S1-IN in
!
interface GigabitEthernet0/0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/0/2
ip address 6.0.0.1 255.255.255.252
!
interface Serial0/1/0
ip address 5.0.0.13 255.255.255.252
!
interface Serial0/1/1
ip address 5.0.0.1 255.255.255.252
```

```
crypto map S1-S2
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
router rip
network 5.0.0.0
network 6.0.0.0
network 10.0.0.0
network 172.20.0.0
!
ip classless
!
ip flow-export version 9
!
!
access-list 101 permit ip 172.20.10.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 101 permit ip 172.20.10.0 0.0.0.255 172.68.0.0 0.0.0.255
access-list 101 permit ip 172.20.10.0 0.0.0.255 198.51.100.0 0.0.0.255
ip access-list extended ACL-S1-IN
permit udp any host 6.0.0.1 eq domain
permit tcp any host 6.0.0.1 eq domain
permit udp host 6.0.0.1 eq domain any
permit tcp host 6.0.0.1 eq domain any
permit tcp any 172.20.10.96 0.0.0.31 eq www
permit tcp any 172.20.10.96 0.0.0.31 eq 443
permit tcp 172.20.10.96 0.0.0.31 eq www any established
permit tcp 172.20.10.96 0.0.0.31 eq 443 any established
permit tcp any 192.168.50.144 0.0.0.7 eq www
permit tcp any 192.168.50.144 0.0.0.7 eq 443
permit tcp 192.168.50.144 0.0.0.7 eq www any established
permit tcp 192.168.50.144 0.0.0.7 eq 443 any established
permit tcp any 172.68.0.96 0.0.0.31 eq www
permit tcp any 172.68.0.96 0.0.0.31 eq 443
permit tcp 172.68.0.96 0.0.0.31 eq www any established
```

```
permit tcp 172.68.0.96 0.0.0.31 eq 443 any established
permit tcp any 198.51.100.96 0.0.0.31 eq www
permit tcp any 198.51.100.96 0.0.0.31 eq 443
permit tcp 198.51.100.96 0.0.0.31 eq www any established
permit tcp 198.51.100.96 0.0.0.31 eq 443 any established
permit ip 172.20.10.0 0.0.0.31 192.168.50.128 0.0.0.15
permit ip 192.168.50.128 0.0.0.15 172.20.10.0 0.0.0.31
permit ip 172.20.10.0 0.0.0.31 172.68.0.0 0.0.0.31
permit ip 172.68.0.0 0.0.0.31 172.20.10.0 0.0.0.31
permit ip 172.20.10.0 0.0.0.31 198.51.100.0 0.0.0.31
permit ip 198.51.100.0 0.0.0.31 172.20.10.0 0.0.0.31
permit ip 172.20.10.32 0.0.0.31 192.168.50.0 0.0.0.63
permit ip 192.168.50.0 0.0.0.63 172.20.10.32 0.0.0.31
permit ip 172.20.10.32 0.0.0.31 172.68.0.32 0.0.0.31
permit ip 172.68.0.32 0.0.0.31 172.20.10.32 0.0.0.31
permit ip 172.20.10.32 0.0.0.31 198.51.100.32 0.0.0.31
permit ip 198.51.100.32 0.0.0.31 172.20.10.32 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 192.168.50.64 0.0.0.31
permit ip 192.168.50.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 172.68.0.64 0.0.0.31
permit ip 172.68.0.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 198.51.100.64 0.0.0.31
permit ip 198.51.100.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 192.168.50.96 0.0.0.31
permit ip 192.168.50.96 0.0.0.31 172.20.10.128 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 172.68.0.128 0.0.0.31
permit ip 172.68.0.128 0.0.0.31 172.20.10.128 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 198.51.100.128 0.0.0.31
permit ip 198.51.100.128 0.0.0.31 172.20.10.128 0.0.0.31
deny ip 172.20.10.0 0.0.0.31 172.20.10.32 0.0.0.31
deny ip 172.20.10.32 0.0.0.31 172.20.10.0 0.0.0.31
deny ip 172.20.10.0 0.0.0.31 172.20.10.64 0.0.0.31
deny ip 172.20.10.64 0.0.0.31 172.20.10.0 0.0.0.31
deny ip 172.20.10.0 0.0.0.31 172.20.10.128 0.0.0.31
deny ip 172.20.10.128 0.0.0.31 172.20.10.0 0.0.0.31
deny ip 172.20.10.32 0.0.0.31 172.20.10.64 0.0.0.31
deny ip 172.20.10.64 0.0.0.31 172.20.10.32 0.0.0.31
deny ip 172.20.10.32 0.0.0.31 172.20.10.128 0.0.0.31
deny ip 172.20.10.64 0.0.0.31 172.20.10.128 0.0.0.31
permit ip any any
!
!
!
!
```

```
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
end
```

```
S1 - S2  
interface Tunnel1  
 ip address 10.0.0.1 255.255.255.252  
 mtu 1476  
 tunnel source Serial0/1/1  
 tunnel destination 5.0.0.2
```

```
S2 - S1  
interface Tunnel1  
 ip address 10.0.0.2 255.255.255.252  
 mtu 1476  
 tunnel source Serial0/1/0  
 tunnel destination 5.0.0.1
```

```
S1 - S3  
interface Tunnel2  
 ip address 10.0.0.5 255.255.255.252  
 mtu 1476  
 tunnel source Gig0/0/2  
 tunnel destination 6.0.0.9
```

```
S3 - S1  
interface Tunnel1  
 ip address 10.0.0.6 255.255.255.252  
 mtu 1476  
 tunnel source Gig0/0/2  
 tunnel destination 6.0.0.1
```

S1 - S4

```
interface Tunnel3
ip address 10.0.0.9 255.255.255.252
mtu 1476
tunnel source Serial0/1/0
tunnel destination 5.0.0.13
```

S4 - S1

```
interface Tunnel3
ip address 10.0.0.10 255.255.255.252
mtu 1476
tunnel source Serial0/1/0
tunnel destination 5.0.0.14
```

S2 - S3

```
interface Tunnel2
ip address 10.0.0.13 255.255.255.252
tunnel source Serial0/1/1
tunnel destination 5.0.0.5
```

S3 - S2

```
interface Tunnel2
ip address 10.0.0.14 255.255.255.252
tunnel source Serial0/1/1
tunnel destination 5.0.0.6
```

S2 - S4

```
interface Tunnel3
ip address 10.0.0.17 255.255.255.252
tunnel source Gig0/0/2
tunnel destination 6.0.0.13
```

S4 - S2

```
interface Tunnel3
ip address 10.0.0.18 255.255.255.252
tunnel source Gig0/0/2
tunnel destination 6.0.0.5
```

S3 - S4

```
interface Tunnel3
ip address 10.0.0.18 255.255.255.252
tunnel source Serial0/1/0
tunnel destination 6.0.0.10
```

S4 - S3

```
interface Tunnel4
ip address 10.0.0.18 255.255.255.252
tunnel source Serial0/1/1
tunnel destination 6.0.0.9
```

ACL SUCURSAL 1

no ip access-list extended ACL-S1-IN

ip access-list extended ACL-S1-IN

!

! ===== 1. PERMITIR DNS =====

permit udp any host 6.0.0.1 eq 53

permit tcp any host 6.0.0.1 eq 53

permit udp host 6.0.0.1 eq 53 any

permit tcp host 6.0.0.1 eq 53 any

!

! ===== 2. PERMITIR HTTP/HTTPS HACIA SERVIDORES WEB =====

! Servidores web S1 (Servers_Net = 172.20.10.96/27)

permit tcp any 172.20.10.96 0.0.0.31 eq 80

permit tcp any 172.20.10.96 0.0.0.31 eq 443

permit tcp 172.20.10.96 0.0.0.31 eq 80 any established

permit tcp 172.20.10.96 0.0.0.31 eq 443 any established

! Servidores web S2 (Servers_Net = 192.168.50.144/29)

permit tcp any 192.168.50.144 0.0.0.7 eq 80

permit tcp any 192.168.50.144 0.0.0.7 eq 443

permit tcp 192.168.50.144 0.0.0.7 eq 80 any established

permit tcp 192.168.50.144 0.0.0.7 eq 443 any established

! Servidores web S3 (Servers_Net = 172.68.0.96/27)

permit tcp any 172.68.0.96 0.0.0.31 eq 80

permit tcp any 172.68.0.96 0.0.0.31 eq 443

permit tcp 172.68.0.96 0.0.0.31 eq 80 any established

permit tcp 172.68.0.96 0.0.0.31 eq 443 any established


```
! Servidores web S4 (Servers_Net = 198.51.100.96/27)
permit tcp any 198.51.100.96 0.0.0.31 eq 80
permit tcp any 198.51.100.96 0.0.0.31 eq 443
permit tcp 198.51.100.96 0.0.0.31 eq 80 any established
permit tcp 198.51.100.96 0.0.0.31 eq 443 any established
!
```

! ===== 3. PERMITIR COMUNICACION MISMA VLAN ENTRE SUCURSALES =====

! VLAN 10 - Mgmt

```
permit ip 172.20.10.0 0.0.0.31 192.168.50.128 0.0.0.15
permit ip 192.168.50.128 0.0.0.15 172.20.10.0 0.0.0.31
permit ip 172.20.10.0 0.0.0.31 172.68.0.0 0.0.0.31
permit ip 172.68.0.0 0.0.0.31 172.20.10.0 0.0.0.31
permit ip 172.20.10.0 0.0.0.31 198.51.100.0 0.0.0.31
permit ip 198.51.100.0 0.0.0.31 172.20.10.0 0.0.0.31
```

! VLAN 20 - Staff

```
permit ip 172.20.10.32 0.0.0.31 192.168.50.0 0.0.0.63
permit ip 192.168.50.0 0.0.0.63 172.20.10.32 0.0.0.31
permit ip 172.20.10.32 0.0.0.31 172.68.0.32 0.0.0.31
permit ip 172.68.0.32 0.0.0.31 172.20.10.32 0.0.0.31
permit ip 172.20.10.32 0.0.0.31 198.51.100.32 0.0.0.31
permit ip 198.51.100.32 0.0.0.31 172.20.10.32 0.0.0.31
```

! VLAN 30 - Wifi

```
permit ip 172.20.10.64 0.0.0.31 192.168.50.64 0.0.0.31
permit ip 192.168.50.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 172.68.0.64 0.0.0.31
permit ip 172.68.0.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 198.51.100.64 0.0.0.31
permit ip 198.51.100.64 0.0.0.31 172.20.10.64 0.0.0.31
```

! VLAN 50 - Guest

```
permit ip 172.20.10.128 0.0.0.31 192.168.50.96 0.0.0.31
permit ip 192.168.50.96 0.0.0.31 172.20.10.128 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 172.68.0.128 0.0.0.31
permit ip 172.68.0.128 0.0.0.31 172.20.10.128 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 198.51.100.128 0.0.0.31
permit ip 198.51.100.128 0.0.0.31 172.20.10.128 0.0.0.31
```

!

! ===== 4. BLOQUEAR COMUNICACION ENTRE VLANs LOCALES =====

! VLAN10 <-> VLAN20

```
deny ip 172.20.10.0 0.0.0.31 172.20.10.32 0.0.0.31
deny ip 172.20.10.32 0.0.0.31 172.20.10.0 0.0.0.31
```

```

! VLAN10 <-> VLAN30
deny ip 172.20.10.0 0.0.0.31 172.20.10.64 0.0.0.31
deny ip 172.20.10.64 0.0.0.31 172.20.10.0 0.0.0.31

! VLAN10 <-> VLAN50
deny ip 172.20.10.0 0.0.0.31 172.20.10.128 0.0.0.31
deny ip 172.20.10.128 0.0.0.31 172.20.10.0 0.0.0.31

! VLAN20 <-> VLAN30
deny ip 172.20.10.32 0.0.0.31 172.20.10.64 0.0.0.31
deny ip 172.20.10.64 0.0.0.31 172.20.10.32 0.0.0.31

! VLAN20 <-> VLAN50
deny ip 172.20.10.32 0.0.0.31 172.20.10.128 0.0.0.31

! VLAN30 <-> VLAN50
deny ip 172.20.10.64 0.0.0.31 172.20.10.128 0.0.0.31
!
! ===== 5. PERMITIR TODO LO DEMÁS =====
permit ip any any

```

ACL SUCURSAL 2

```

no ip access-list extended ACL-S2-IN
ip access-list extended ACL-S2-IN
!
! ===== 1. PERMITIR DNS =====
permit udp any host 6.0.0.5 eq 53
permit tcp any host 6.0.0.5 eq 53
permit udp host 6.0.0.5 eq 53 any
permit tcp host 6.0.0.5 eq 53 any
!
! ===== 2. PERMITIR HTTP/HTTPS HACIA SERVIDORES WEB =====
! Servidores web S1 (Servers_Net = 172.20.10.96/27)
permit tcp any 172.20.10.96 0.0.0.31 eq 80
permit tcp any 172.20.10.96 0.0.0.31 eq 443
permit tcp 172.20.10.96 0.0.0.31 eq 80 any established
permit tcp 172.20.10.96 0.0.0.31 eq 443 any established
!
! Servidores web S2 (Servers_Net = 192.168.50.144/29)
permit tcp any 192.168.50.144 0.0.0.7 eq 80
permit tcp any 192.168.50.144 0.0.0.7 eq 443
permit tcp 192.168.50.144 0.0.0.7 eq 80 any established
permit tcp 192.168.50.144 0.0.0.7 eq 443 any established

```

!

! Servidores web S3 (Servers_Net = 172.68.0.96/27)

permit tcp any 172.68.0.96 0.0.0.31 eq 80

permit tcp any 172.68.0.96 0.0.0.31 eq 443

permit tcp 172.68.0.96 0.0.0.31 eq 80 any established

permit tcp 172.68.0.96 0.0.0.31 eq 443 any established

!

! Servidores web S4 (Servers_Net = 198.51.100.96/27)

permit tcp any 198.51.100.96 0.0.0.31 eq 80

permit tcp any 198.51.100.96 0.0.0.31 eq 443

permit tcp 198.51.100.96 0.0.0.31 eq 80 any established

permit tcp 198.51.100.96 0.0.0.31 eq 443 any established

!

! ===== 3. PERMITIR COMUNICACIÓN MISMA VLAN ENTRE SUCURSALES =====

! VLAN 10 - Management

permit ip 192.168.50.128 0.0.0.15 172.20.10.0 0.0.0.31

permit ip 172.20.10.0 0.0.0.31 192.168.50.128 0.0.0.15

permit ip 192.168.50.128 0.0.0.15 172.68.0.0 0.0.0.31

permit ip 172.68.0.0 0.0.0.31 192.168.50.128 0.0.0.15

permit ip 192.168.50.128 0.0.0.15 198.51.100.0 0.0.0.31

permit ip 198.51.100.0 0.0.0.31 192.168.50.128 0.0.0.15

!

! VLAN 20 - Staff

permit ip 192.168.50.0 0.0.0.63 172.20.10.32 0.0.0.31

permit ip 172.20.10.32 0.0.0.31 192.168.50.0 0.0.0.63

permit ip 192.168.50.0 0.0.0.63 172.68.0.32 0.0.0.31

permit ip 172.68.0.32 0.0.0.31 192.168.50.0 0.0.0.63

permit ip 192.168.50.0 0.0.0.63 198.51.100.32 0.0.0.31

permit ip 198.51.100.32 0.0.0.31 192.168.50.0 0.0.0.63

!

! VLAN 30 - Wifi

permit ip 192.168.50.64 0.0.0.31 172.20.10.64 0.0.0.31

permit ip 172.20.10.64 0.0.0.31 192.168.50.64 0.0.0.31

permit ip 192.168.50.64 0.0.0.31 172.68.0.64 0.0.0.31

permit ip 172.68.0.64 0.0.0.31 192.168.50.64 0.0.0.31

permit ip 192.168.50.64 0.0.0.31 198.51.100.64 0.0.0.31

permit ip 198.51.100.64 0.0.0.31 192.168.50.64 0.0.0.31

!

! VLAN 50 - Guest

permit ip 192.168.50.96 0.0.0.31 172.20.10.128 0.0.0.31

permit ip 172.20.10.128 0.0.0.31 192.168.50.96 0.0.0.31

permit ip 192.168.50.96 0.0.0.31 172.68.0.128 0.0.0.31

permit ip 172.68.0.128 0.0.0.31 192.168.50.96 0.0.0.31

permit ip 192.168.50.96 0.0.0.31 198.51.100.128 0.0.0.31

```

permit ip 198.51.100.128 0.0.0.31 192.168.50.96 0.0.0.31
!
! ===== 4. BLOQUEAR COMUNICACIÓN ENTRE VLANs LOCALES =====
! VLAN10 <-> VLAN20
deny ip 192.168.50.128 0.0.0.15 192.168.50.0 0.0.0.63
deny ip 192.168.50.0 0.0.0.63 192.168.50.128 0.0.0.15
!
! VLAN10 <-> VLAN30
deny ip 192.168.50.128 0.0.0.15 192.168.50.64 0.0.0.31
deny ip 192.168.50.64 0.0.0.31 192.168.50.128 0.0.0.15
!
! VLAN10 <-> VLAN50
deny ip 192.168.50.128 0.0.0.15 192.168.50.96 0.0.0.31
deny ip 192.168.50.96 0.0.0.31 192.168.50.128 0.0.0.15
!
! VLAN20 <-> VLAN30
deny ip 192.168.50.0 0.0.0.63 192.168.50.64 0.0.0.31
deny ip 192.168.50.64 0.0.0.31 192.168.50.0 0.0.0.63
!
! VLAN20 <-> VLAN50
deny ip 192.168.50.0 0.0.0.63 192.168.50.96 0.0.0.31
deny ip 192.168.50.96 0.0.0.31 192.168.50.0 0.0.0.63
!
! VLAN30 <-> VLAN50
deny ip 192.168.50.64 0.0.0.31 192.168.50.96 0.0.0.31
deny ip 192.168.50.96 0.0.0.31 192.168.50.64 0.0.0.31
!
! ===== 5. PERMITIR TODO LO DEMÁS =====
permit ip any any

```

ACL SUCURSAL 3

```

no ip access-list extended ACL-S3-IN
ip access-list extended ACL-S3-IN
!
! ===== 1. PERMITIR DNS =====
permit udp any host 6.0.0.9 eq 53
permit tcp any host 6.0.0.9 eq 53
permit udp host 6.0.0.9 eq 53 any
permit tcp host 6.0.0.9 eq 53 any
!
! ===== 2. PERMITIR HTTP/HTTPS HACIA SERVIDORES WEB =====
! Servidores web S1 (172.20.10.96/27)
permit tcp any 172.20.10.96 0.0.0.31 eq 80

```

```
permit tcp any 172.20.10.96 0.0.0.31 eq 443
permit tcp 172.20.10.96 0.0.0.31 eq 80 any established
permit tcp 172.20.10.96 0.0.0.31 eq 443 any established
```

!

! Servidores web S2 (192.168.50.144/29)

```
permit tcp any 192.168.50.144 0.0.0.7 eq 80
permit tcp any 192.168.50.144 0.0.0.7 eq 443
permit tcp 192.168.50.144 0.0.0.7 eq 80 any established
permit tcp 192.168.50.144 0.0.0.7 eq 443 any established
```

!

! Servidores web S3 (172.68.0.96/27)

```
permit tcp any 172.68.0.96 0.0.0.31 eq 80
permit tcp any 172.68.0.96 0.0.0.31 eq 443
permit tcp 172.68.0.96 0.0.0.31 eq 80 any established
permit tcp 172.68.0.96 0.0.0.31 eq 443 any established
```

!

! Servidores web S4 (198.51.100.96/27)

```
permit tcp any 198.51.100.96 0.0.0.31 eq 80
permit tcp any 198.51.100.96 0.0.0.31 eq 443
permit tcp 198.51.100.96 0.0.0.31 eq 80 any established
permit tcp 198.51.100.96 0.0.0.31 eq 443 any established
```

!

! ===== 3. PERMITIR COMUNICACIÓN MISMA VLAN ENTRE SUCURSALES =====

! VLAN 10 - Management

```
permit ip 172.68.0.0 0.0.0.31 172.20.10.0 0.0.0.31
permit ip 172.20.10.0 0.0.0.31 172.68.0.0 0.0.0.31
permit ip 172.68.0.0 0.0.0.31 192.168.50.128 0.0.0.15
permit ip 192.168.50.128 0.0.0.15 172.68.0.0 0.0.0.31
permit ip 172.68.0.0 0.0.0.31 198.51.100.0 0.0.0.31
permit ip 198.51.100.0 0.0.0.31 172.68.0.0 0.0.0.31
```

!

! VLAN 20 - Staff

```
permit ip 172.68.0.32 0.0.0.31 172.20.10.32 0.0.0.31
permit ip 172.20.10.32 0.0.0.31 172.68.0.32 0.0.0.31
permit ip 172.68.0.32 0.0.0.31 192.168.50.0 0.0.0.63
permit ip 192.168.50.0 0.0.0.63 172.68.0.32 0.0.0.31
permit ip 172.68.0.32 0.0.0.31 198.51.100.32 0.0.0.31
permit ip 198.51.100.32 0.0.0.31 172.68.0.32 0.0.0.31
```

!

! VLAN 30 - Wifi

```
permit ip 172.68.0.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 172.68.0.64 0.0.0.31
permit ip 172.68.0.64 0.0.0.31 192.168.50.64 0.0.0.31
permit ip 192.168.50.64 0.0.0.31 172.68.0.64 0.0.0.31
```

```

permit ip 172.68.0.64 0.0.0.31 198.51.100.64 0.0.0.31
permit ip 198.51.100.64 0.0.0.31 172.68.0.64 0.0.0.31
!
! VLAN 50 - Guest
permit ip 172.68.0.128 0.0.0.31 172.20.10.128 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 172.68.0.128 0.0.0.31
permit ip 172.68.0.128 0.0.0.31 192.168.50.96 0.0.0.31
permit ip 192.168.50.96 0.0.0.31 172.68.0.128 0.0.0.31
permit ip 172.68.0.128 0.0.0.31 198.51.100.128 0.0.0.31
permit ip 198.51.100.128 0.0.0.31 172.68.0.128 0.0.0.31
!
! ===== 4. BLOQUEAR COMUNICACIÓN ENTRE VLANs LOCALES =====
! VLAN10 <-> VLAN20
deny ip 172.68.0.0 0.0.0.31 172.68.0.32 0.0.0.31
deny ip 172.68.0.32 0.0.0.31 172.68.0.0 0.0.0.31
!
! VLAN10 <-> VLAN30
deny ip 172.68.0.0 0.0.0.31 172.68.0.64 0.0.0.31
deny ip 172.68.0.64 0.0.0.31 172.68.0.0 0.0.0.31
!
! VLAN10 <-> VLAN50
deny ip 172.68.0.0 0.0.0.31 172.68.0.128 0.0.0.31
deny ip 172.68.0.128 0.0.0.31 172.68.0.0 0.0.0.31
!
! VLAN20 <-> VLAN30
deny ip 172.68.0.32 0.0.0.31 172.68.0.64 0.0.0.31
deny ip 172.68.0.64 0.0.0.31 172.68.0.32 0.0.0.31
!
! VLAN20 <-> VLAN50
deny ip 172.68.0.32 0.0.0.31 172.68.0.128 0.0.0.31
deny ip 172.68.0.128 0.0.0.31 172.68.0.32 0.0.0.31
!
! VLAN30 <-> VLAN50
deny ip 172.68.0.64 0.0.0.31 172.68.0.128 0.0.0.31
deny ip 172.68.0.128 0.0.0.31 172.68.0.64 0.0.0.31
!
! ===== 5. PERMITIR TODO LO DEMÁS =====
permit ip any any

```

ACL SUCURSAL 4

```

no ip access-list extended ACL-S4-IN
ip access-list extended ACL-S4-IN
!

```

! ===== 1. PERMITIR DNS =====

permit udp any host 6.0.0.13 eq 53

permit tcp any host 6.0.0.13 eq 53

permit udp host 6.0.0.13 eq 53 any

permit tcp host 6.0.0.13 eq 53 any

!

! ===== 2. PERMITIR HTTP/HTTPS HACIA SERVIDORES WEB =====

! Servidores S1 (172.20.10.96/27)

permit tcp any 172.20.10.96 0.0.0.31 eq 80

permit tcp any 172.20.10.96 0.0.0.31 eq 443

permit tcp 172.20.10.96 0.0.0.31 eq 80 any established

permit tcp 172.20.10.96 0.0.0.31 eq 443 any established

!

! Servidores S2 (192.168.50.144/29)

permit tcp any 192.168.50.144 0.0.0.7 eq 80

permit tcp any 192.168.50.144 0.0.0.7 eq 443

permit tcp 192.168.50.144 0.0.0.7 eq 80 any established

permit tcp 192.168.50.144 0.0.0.7 eq 443 any established

!

! Servidores S3 (172.68.0.96/27)

permit tcp any 172.68.0.96 0.0.0.31 eq 80

permit tcp any 172.68.0.96 0.0.0.31 eq 443

permit tcp 172.68.0.96 0.0.0.31 eq 80 any established

permit tcp 172.68.0.96 0.0.0.31 eq 443 any established

!

! Servidores S4 (198.51.100.96/27)

permit tcp any 198.51.100.96 0.0.0.31 eq 80

permit tcp any 198.51.100.96 0.0.0.31 eq 443

permit tcp 198.51.100.96 0.0.0.31 eq 80 any established

permit tcp 198.51.100.96 0.0.0.31 eq 443 any established

!

! ===== 3. PERMITIR COMUNICACIÓN MISMA VLAN ENTRE SUCURSALES =====

! VLAN 10 - Management

permit ip 198.51.100.0 0.0.0.31 172.20.10.0 0.0.0.31

permit ip 172.20.10.0 0.0.0.31 198.51.100.0 0.0.0.31

permit ip 198.51.100.0 0.0.0.31 192.168.50.128 0.0.0.15

permit ip 192.168.50.128 0.0.0.15 198.51.100.0 0.0.0.31

permit ip 198.51.100.0 0.0.0.31 172.68.0.0 0.0.0.31

permit ip 172.68.0.0 0.0.0.31 198.51.100.0 0.0.0.31

!

! VLAN 20 - Staff

permit ip 198.51.100.32 0.0.0.31 172.20.10.32 0.0.0.31

permit ip 172.20.10.32 0.0.0.31 198.51.100.32 0.0.0.31

permit ip 198.51.100.32 0.0.0.31 192.168.50.0 0.0.0.63

```
permit ip 192.168.50.0 0.0.0.63 198.51.100.32 0.0.0.31
permit ip 198.51.100.32 0.0.0.31 172.68.0.32 0.0.0.31
permit ip 172.68.0.32 0.0.0.31 198.51.100.32 0.0.0.31
!
```

! VLAN 30 - Wifi

```
permit ip 198.51.100.64 0.0.0.31 172.20.10.64 0.0.0.31
permit ip 172.20.10.64 0.0.0.31 198.51.100.64 0.0.0.31
permit ip 198.51.100.64 0.0.0.31 192.168.50.64 0.0.0.31
permit ip 192.168.50.64 0.0.0.31 198.51.100.64 0.0.0.31
permit ip 198.51.100.64 0.0.0.31 172.68.0.64 0.0.0.31
permit ip 172.68.0.64 0.0.0.31 198.51.100.64 0.0.0.31
!
```

! VLAN 50 - Guest

```
permit ip 198.51.100.128 0.0.0.31 172.20.10.128 0.0.0.31
permit ip 172.20.10.128 0.0.0.31 198.51.100.128 0.0.0.31
permit ip 198.51.100.128 0.0.0.31 192.168.50.96 0.0.0.31
permit ip 192.168.50.96 0.0.0.31 198.51.100.128 0.0.0.31
permit ip 198.51.100.128 0.0.0.31 172.68.0.128 0.0.0.31
permit ip 172.68.0.128 0.0.0.31 198.51.100.128 0.0.0.31
!
```

! ===== 4. BLOQUEAR COMUNICACIÓN ENTRE VLANs LOCALES =====

! VLAN10 <-> VLAN20

```
deny ip 198.51.100.0 0.0.0.31 198.51.100.32 0.0.0.31
deny ip 198.51.100.32 0.0.0.31 198.51.100.0 0.0.0.31
!
```

! VLAN10 <-> VLAN30

```
deny ip 198.51.100.0 0.0.0.31 198.51.100.64 0.0.0.31
deny ip 198.51.100.64 0.0.0.31 198.51.100.0 0.0.0.31
!
```

! VLAN10 <-> VLAN50

```
deny ip 198.51.100.0 0.0.0.31 198.51.100.128 0.0.0.31
deny ip 198.51.100.128 0.0.0.31 198.51.100.0 0.0.0.31
!
```

! VLAN20 <-> VLAN30

```
deny ip 198.51.100.32 0.0.0.31 198.51.100.64 0.0.0.31
deny ip 198.51.100.64 0.0.0.31 198.51.100.32 0.0.0.31
!
```

! VLAN20 <-> VLAN50

```
deny ip 198.51.100.32 0.0.0.31 198.51.100.128 0.0.0.31
deny ip 198.51.100.128 0.0.0.31 198.51.100.32 0.0.0.31
!
```

! VLAN30 <-> VLAN50

```
deny ip 198.51.100.64 0.0.0.31 198.51.100.128 0.0.0.31
deny ip 198.51.100.128 0.0.0.31 198.51.100.64 0.0.0.31
```


!

!===== 5. PERMITIR TODO LO DEMÁS =====

permit ip any any